

bm

Project Report

1. Readme

badmonkey

This is a clone of the digitech Bad Monkey overdrive. This is apparently a derivative of the tubescreamer pedal from some time ago. This uses a gyrator circuit for the low's.

Very nice sound.

This is the 2nd version of the pedal. The first version was missing the R15/C12. This was added to the first version of the pedal as a bodge.

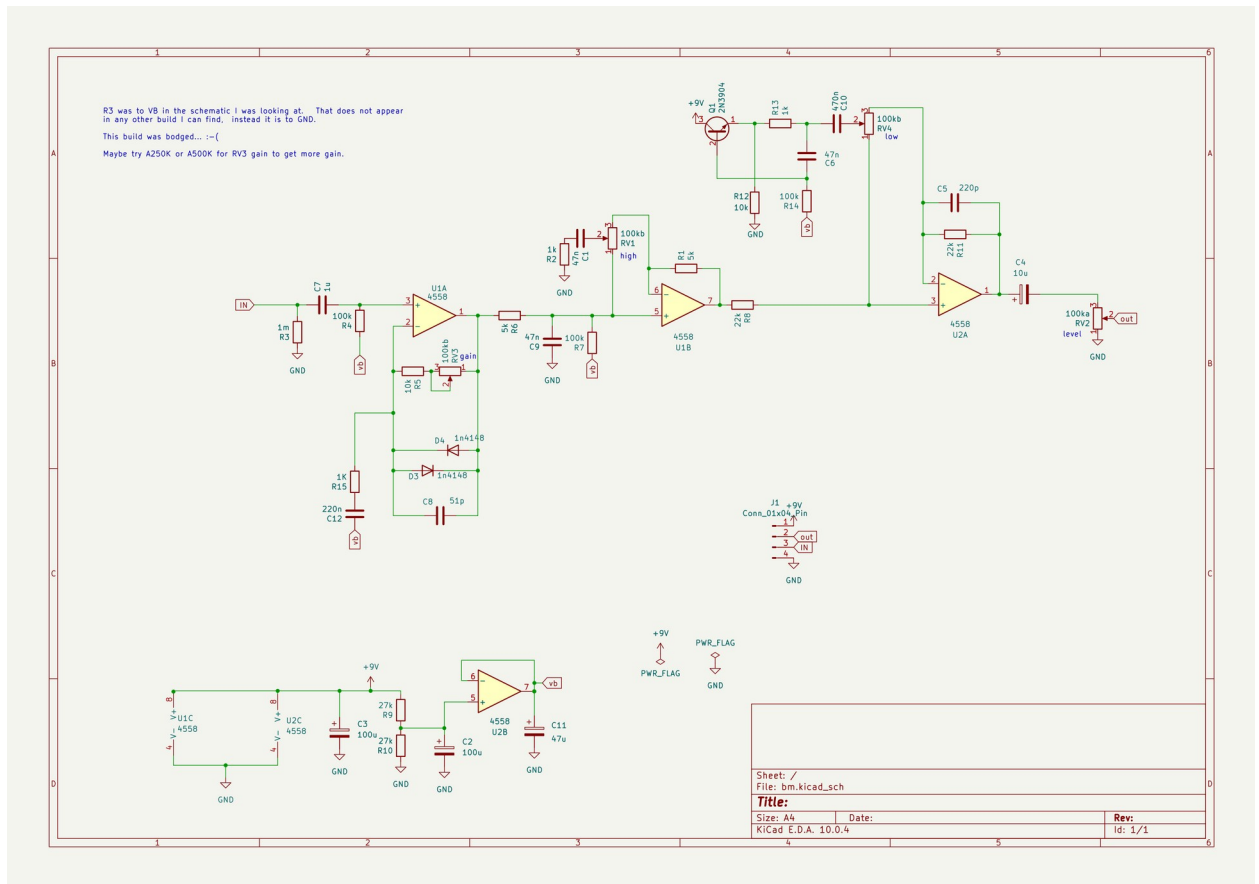
Third version being ordered now..

V2 was slightly too large to squeeze into 125B, so adjusted it to 59.5mm wide.

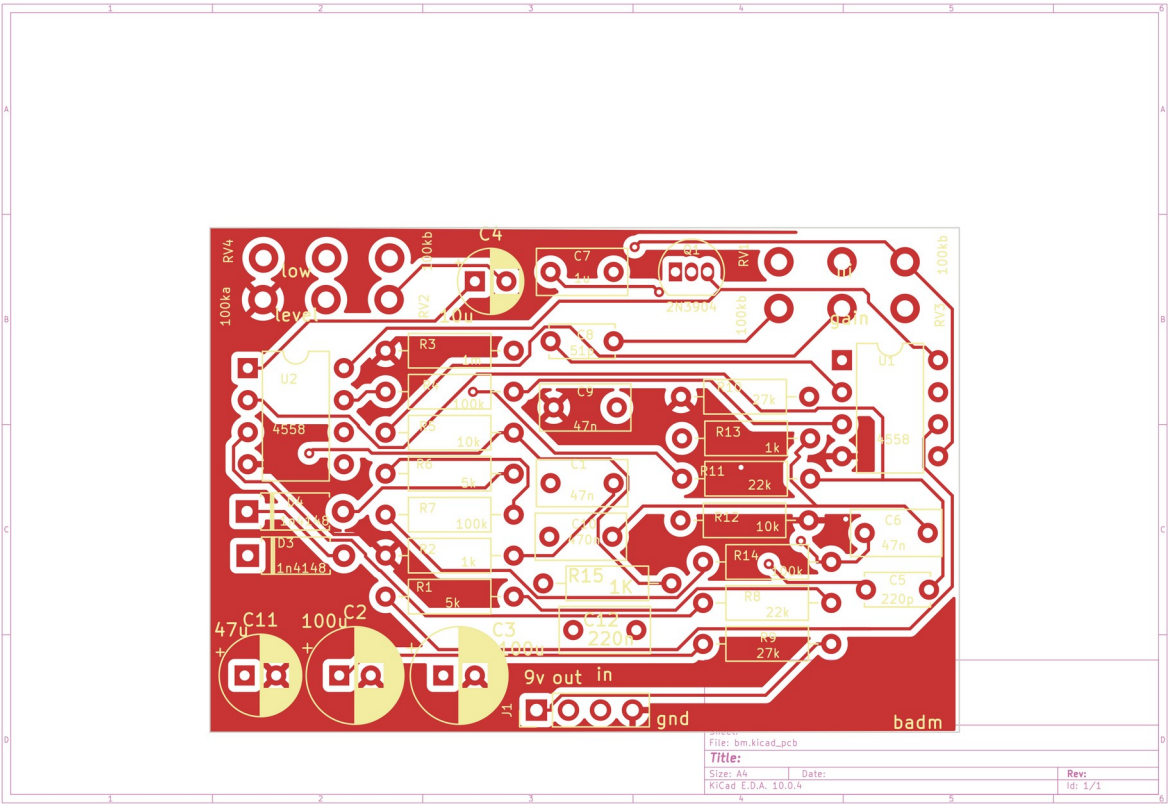
So this pedal circuit is verified. I have not boxed this one yet, either.

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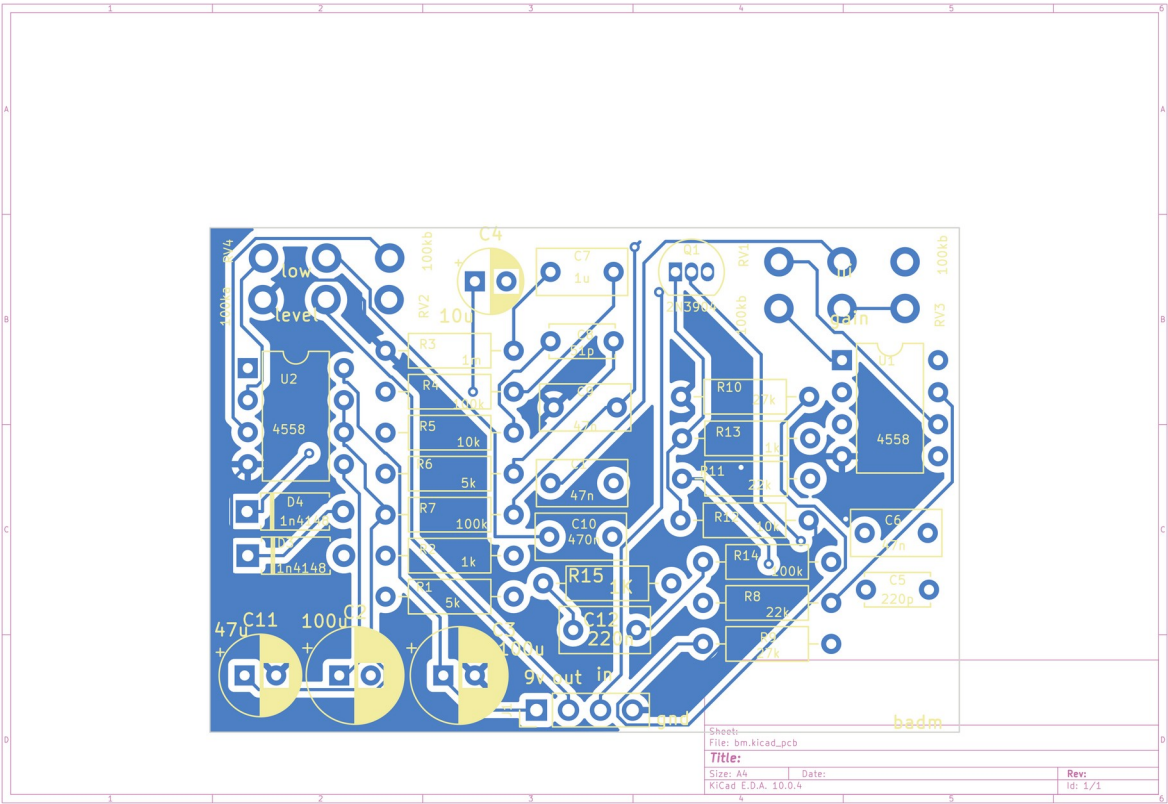
2. Schematic



3. PCB F.Cu + Silkscreen



4. PCB B.CU + Silkscreen



5. Bill of Materials

Reference	Value	Footprint	Qty	DNP
C1,C6,C9	47n	Capacitor_THT :C_Rect_L7.0m m_W3.5mm_P 5.00mm	3	
C2,C3	100u	Capacitor_THT :CP_Radial_D7. 5mm_P2.50m m	2	
C4	10u	Capacitor_THT :CP_Radial_D5. 0mm_P2.50m m	1	
C5	220p	Capacitor_THT :C_Disc_D5.0m m_W2.5mm_P 5.00mm	1	
C7	1u	Capacitor_THT :C_Rect_L7.0m m_W3.5mm_P 5.00mm	1	
C8	51p	Capacitor_THT :C_Disc_D5.0m m_W2.5mm_P 5.00mm	1	
C10	470n	Capacitor_THT :C_Rect_L7.0m m_W3.5mm_P 5.00mm	1	
C11	47u	Capacitor_THT :CP_Radial_D6. 3mm_P2.50m m	1	
C12	220n	Capacitor_THT :C_Rect_L7.0m m_W3.5mm_P 5.00mm	1	
D3,D4	1n4148	Diode_THT:D_ A- 405_P7.62mm	2	

		_Horizontal		
J1	Conn_01x04_Pin	Connector_Pin Header_2.54mm:PinHeader_1x04_P2.54mm_Vertical	1	
Q1	2N3904	Package_TO_SOT_THT:TO-92_Inline	1	
R1,R6	5k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal	2	
R2,R13	1k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal	2	
R3	1m	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal	1	
R4,R7,R14	100k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal	3	
R5,R12	10k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal	2	
R8,R11	22k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D	2	

		2.5mm_P10.1 6mm_Horizon tal		
R9,R10	27k	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizon tal	2	
R15	1K	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizon tal	1	
RV1,RV3,RV4	100kb	pedals:Pot_Un derside	3	
RV2	100ka	pedals:Pot_Un derside	1	
U1,U2	4558	Package_DIP: DIP- 8_W7.62mm	2	

Total line items: 23